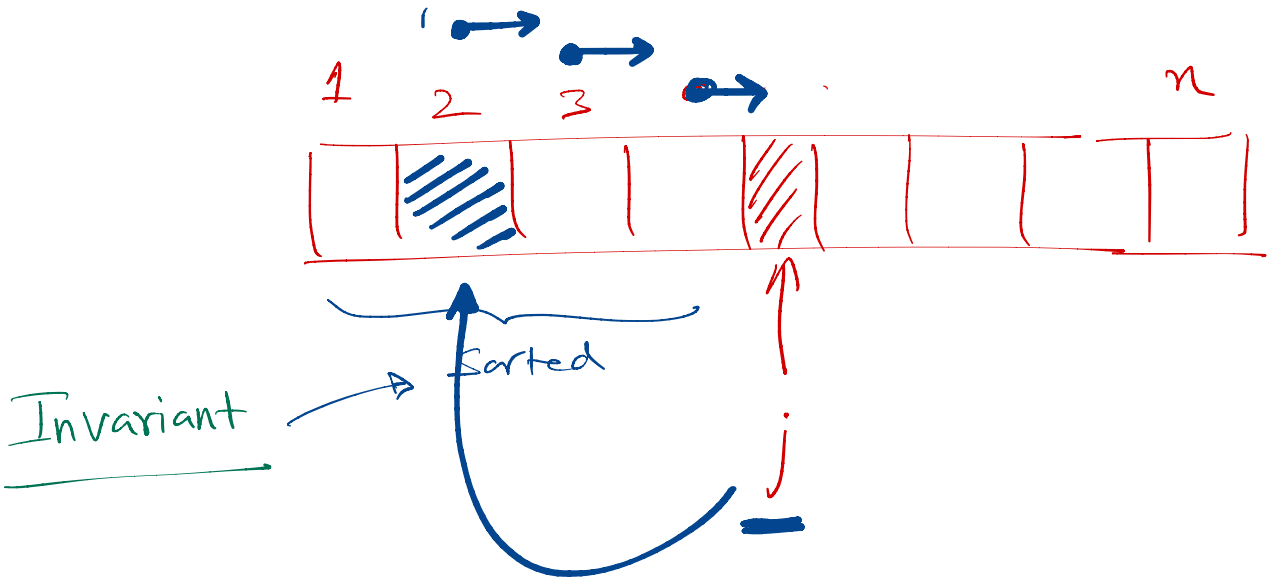


$$T(n) = a n^3 + b n^2 + c n + d$$

machine dependent const.

Study $T(n)$ in the limit $n \to \infty$

Asymptotic analysis

$T(n) = \Theta(n^3) \rightarrow$ Growth

$Q \quad f(n) = n^2$

$c \quad g(n) = n^3$

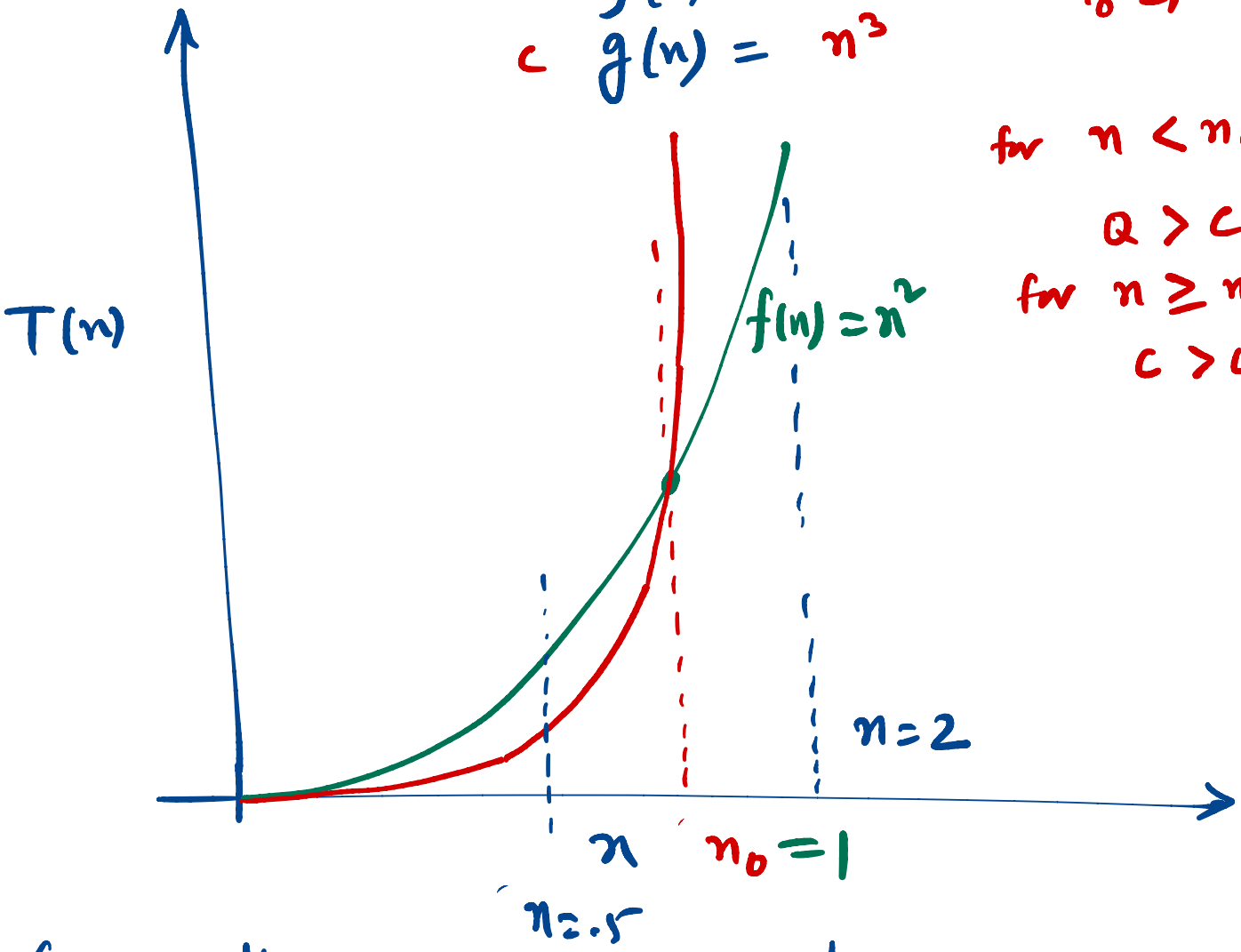
$n_0 = 1$

for $n < n_0$

$Q > c$

for $n \geq n_0$

$c > Q$



$f(.5) = 1/4$

$g(.5) = 1/8$

$\Rightarrow f(.5) > g(.5)$

$f(2) = 4$

$g(2) = 8$

$\Rightarrow f(2) < g(2)$